

Michael Schröder

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Experience

- 2019–present **Research Assistant**, *TU Wien*, Vienna, Austria.
Graduate research in programming languages and software engineering.
Currently working on refinement type systems for ad hoc grammar inference.
Funded in part by the Austrian Science Fund (FWF) [10.55776/PIN3275223].
- Aug–Nov 2021 **Research Internship**, *Microsoft Research*, Cambridge, UK.
Worked with Katja Kevic and Brendan Murphy to find feature flag interdependencies in the Microsoft Office codebase. Developed causal inference method based on patterns in telemetry data. A paper based on this work was published at ESEC/FSE 2022.
- Oct–Nov 2019 **Research Visit**, *MIT CSAIL*, Cambridge, MA.
Hosted by Jürgen Cito and Martin Rinard. Worked on large-scale analysis of personal configuration scripts on GitHub. Resulted in publication of paper in EMSE.
Funded by the Christiana Hörbiger award for the mobility of young scientists.
- 2016–2017 **Co-Founder & CTO**, *Auctionist*, Vienna, Austria.
An online auction house for fine art and collectibles. Business development, interaction design, full-stack web engineering. Managed scalable deployments of Haskell backend.
- 2015–2021 **Mobile App Developer / AR Consultant**, *Refrakt*, Berlin, Germany.
Developed artistic, educational, and commercial mixed reality experiences, including:
— the *Refrakt* app, a virtual museum in augmented reality,
— *SwellAR*, a prototype visualising real-time oceanic currents for the Humboldt Forum,
— *TAMAM*, a companion app for Berlin’s Museum of Islamic Art.
- 2015–2017 **University Assistant**, *TU Wien*, Vienna, Austria.
Taught programming to ~800 undergraduate computer science students per year.
Responsible for weekly recitations in smaller groups and devising and grading of exams.
- Mar–Jun 2014 **Research Practicum**, *University of New South Wales (UNSW)*, Sydney, Australia.
Developed *Transactional Memory with Finalizers*, an extension to the runtime system of the Glasgow Haskell Compiler (GHC). Invited to present work at Oracle Labs, Brisbane.
Funded by a stipend from TU Wien.

Education

- 2020–present **PhD, Computer Science**, *TU Wien*, Vienna, Austria.
Working on inferring formal grammars for ad hoc parsers via refinement types.
Advised by Jürgen Cito.
- 2012–2015 **MSc, Software Engineering & Internet Computing**, *TU Wien*, Vienna, Austria.
Thesis: *Durability and Contention in Software Transactional Memory*.
Advised by Jens Knoop and Gabriele Keller (UNSW). *Graduated with distinction.*
- 2007–2012 **BSc, Software & Information Engineering**, *TU Wien*, Vienna, Austria.
Thesis: *Optimizing Lua using run-time type specialization*.
Advised by M. Anton Ertl. *Graduated with distinction.*

Selected Publications

- OOPSLA **Static Inference of Regular Grammars for Ad Hoc Parsers.**
2025 Michael Schröder and Jürgen Cito.
- ESEC/FSE **Discovering Feature Flag Interdependencies in Microsoft Office.**
2022 Michael Schröder, Katja Kevic, Dan Gopstein, Brendan Murphy, Jennifer Beckmann.
- ICSE NIER **Grammars for Free: Toward Grammar Inference for Ad Hoc Parsers.**
2022 Michael Schröder and Jürgen Cito.
- EMSE **An Empirical Investigation of Command-Line Customization.**
2022 Michael Schröder and Jürgen Cito.

Service

I am artifact evaluation co-chair of **The Programming Journal** (Volume 11).

I have served on the program committee of the **FSE** 2026 Ideas, Visions, and Reflections track; the junior program committee for **MSR** (2023–24); the extended review committee for **ECOOP** (2022–23); the artifact evaluation committees for **PLDI** (2023–24), **ESOP/FASE/FoSSaCS** (2024), **ECOOP** (2022–23), **ISSTA** (2022–24), and **The Programming Journal** (Volumes 7–10); the program committee of the **CIKM** 2020 resource track; and was an additional reviewer for the **ICSE** 2021 Demo track.

I was a beginner's mentor at **ZuriHac** (2015) and served as a student volunteer at the **Vienna Summer of Logic** (2014) and **HiPEAC** (2014).

Teaching

- 2019–2026 **Web Engineering**, *TU Wien*.
— winner of the *Best Distance Learning Award* (2020).
— finalist for *Best Lecture Award* (2022) and *Special Award for Digital Teaching* (2022).
- 2015–2017 **Program Construction**, *TU Wien*.
- 2012–2014 **Functional Programming** (Student Tutor), *TU Wien*.

Other Projects

- 2015–present **Kürzestfilm Festspiele**, Vienna, Austria.
The world's shortest film festival, where every film has to be exactly twelve seconds long. Hosting and co-writing the annual award show and related events.
- 2015 **Queerbot**, *Roboexotica*, Vienna, Austria.
A cocktail robot that aimed to get people drunk and challenge misconceptions about genotype and phenotypic expression of gender and sexuality. Coding and UX design.

Language Skills

- English (fluent), German (native).
- Haskell, Python, C, Obj-C, Swift, JavaScript, Java, SQL.